

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	03 November 2025
Team ID	NM2025TMID03627
Project Name	To Supply Leftover Food to Poor
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Donor Registration	Registration through Form, Registration through Gmail, Registration through LinkedIn
FR-2	Donor Confirmation	Confirmation via Email, Confirmation via OTP
FR-3	Food Donation Listing	Donors can list available leftover food, Include food type, quantity, and pickup location, Specify expiry time/freshness duration
FR-4	Volunteer/NGO Registration	Registration through Form, Verification process for NGOs, Volunteer background check integration
FR-5	Food Availability Check	System displays available food donations in real-time, Filter by location, food type, and quantity, Show distance from volunteer/NGO location
FR-6	Food Request & Assignment	Volunteers/NGOs can request food donations, System assigns food to nearest available volunteer, Send pickup confirmation to donor
FR-7	Pickup Scheduling	Schedule pickup time with donor, Send reminders to both parties, Track pickup status (Pending, In Transit, Completed)
FR-8	Distribution Tracking	Record distribution location and beneficiary count, Upload photos as proof of distribution, Mark donation as completed
FR-9	Notification System	Real-time alerts for new food donations, Pickup reminders, Status update notifications
FR-10	Reporting & Analytics	Generate reports on food collected and distributed, Track donor contributions, Monitor volunteer/NGO performance, Visualize impact metrics on dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be intuitive for donors, volunteers, and NGOs with minimal training required.
NFR-2	Security	Role-based access control for donors, volunteers, NGOs, and admins. Secure storage of personal and location data. Authentication and authorization protocols.
NFR-3	Reliability	The system must accurately track food availability and assignments. Ensure data consistency across all transactions. 99.5% uptime requirement.
NFR-4	Performance	Food listing should appear in real-time. Search and filter operations should complete within 2 seconds. Support concurrent users without performance degradation.
NFR-5	Availability	The system should be accessible 24/7 for emergency food donations. Mobile-responsive design for on-the-go access.
NFR-6	Scalability	The system should handle increasing numbers of donors, volunteers, and food listings. Support multiple cities and regions. Scale to accommodate growing NGO partnerships.
NFR-7	Transparency	Complete audit trail of food donations from listing to distribution. Transparent reporting for stakeholders. Public dashboard showing impact metrics.
NFR-8	Integration	Integrate with Google Maps for location services. SMS/Email gateway integration for notifications. Salesforce CRM integration for data management.
NFR-9	Compliance	Adhere to food safety and handling guidelines. Comply with data privacy regulations (GDPR, local laws). Maintain records for regulatory reporting.
NFR-10	Maintainability	Modular design for easy updates and feature additions. Comprehensive documentation for system maintenance. Regular backup and disaster recovery procedures.